Preliminary Design Review (PDR) Report

Project: MedicalAlert Wristband **Team:** Eyobed, Issac, Asrael, Mohammad
Submission Date: March 3, 2026 **Team Name:** Nexus

1. Goals

The goal is to design a wrist-worn medication reminder to assist elderly patients in keeping up with their prescribed medication schedules. The device specifically targets patients prescribed Lisinopril, one of the most common antihypertensives (blood pressure medications) in the United States. Missing these medications is a leading cause of preventable cardiovascular issues in older adults. This device directly addresses that risk by providing a reliable, wrist-based alert system. The device alerts the user through vibration at the scheduled medication time and requires an explicit button press to acknowledge the alert, reducing reliance on memory. It is intended to be distributed directly by pharmacists alongside the patient's prescription, with setup assistance from caregivers or family when needed

2. Scope

The scope of this project covers the design, prototyping, and testing of the MedicalAlert Wristband through the PDR and CDR phases. The prototype consists of:

- A wrist-mounted 3D-printed enclosure (nominally 55 × 25 × 15 mm) with attachment points for standard silicone watch straps.
- An Arduino Nano 33 BLE Sense Rev2 microcontroller as the main controller.
- A coin vibration motor driven through a transistor and flyback diode for reliable haptic alerts.
- A tactile push button on the top surface to allow the user to dismiss alerts with a single press.
- A single-cell rechargeable Li-Po battery and a TP4056-based charging module, enabling recharging via a USB connector and providing several days of operation between charges.
- Simple internal software timekeeping based on the microcontroller's clock to schedule a daily alert time.

3. Stakeholders

Stakeholder	Role / Interest
Elderly patients (60+)	benefit from medication adherence and reduced burden to remember to take it
Pharmacists	Device distributors who configure the initial medication schedule
Caregivers/ Family	assist with setup and monitoring

4. Schedule (Gantt Chart) and Group Tasks

Phase	Task	Lead	Start	End	Days
Phase 1: Requirements & Planning	Team Formation & Role Assignment	All	2/2/26	2/8/26	7
	Define Scope	Issac	2/9/26	2/22/26	14
	Define Stakeholders	Eyobed	2/9/26	2/22/26	14
	Define Requirements	Mohammed	2/9/26	2/22/26	14
	Define TPMs & KPP	Asrael	2/16/26	2/22/26	7
	Define Initial Schedule	Asrael	2/17/26	2/23/26	7
	Write & Compile PDR Report	Issac	2/23/26	3/2/26	8
	Submit PDR Report *(milestone)*	Eyobed	**3/3/26**	**3/3/26**	1
Phase 2: System Design	Hardware Design	Eyobed	3/4/26	3/18/26	15
	Mechanical Design	Asrael	3/4/26	3/18/26	15
	Software Design	Issac	3/4/26	3/18/26	15
	Finalize Parts List	Eyobed	3/15/26	3/24/26	10
	Order Components	Mohammad	3/24/26	3/25/26	2
	Peer Review *(milestone)*	All	**3/26/26**	**3/26/26**	1
Phase 3: Implementation & Prototyping	Breadboard Prototype	Asrael	3/28/26	4/6/26	10
	Write Code	Asrael	3/28/26	4/11/26	15
	Construct Enclosure	Eyobed	3/30/26	4/8/26	10
	Initial Assembly	Issac	4/9/26	4/13/26	5
	Write & Compile CDR Report	Issac	4/7/26	4/13/26	7
	Submit CDR Report *(milestone)*	Mohammad	**4/14/26**	**4/14/26**	1
Phase 4: Testing & Integration	TPM 1 Test	All	4/14/26	4/16/26	3
	TPM 2 Test	All	4/14/26	4/16/26	3
	KPP Test	All	4/14/26	4/16/26	3
	Integration & Bug Fixing	Issac, Asrael	4/17/26	4/19/26	3
Phase 5: Final Presentation & Reporting	Prepare Presentation	All	4/19/26	4/22/26	4
	Rehearse Presentation	All	4/19/26	4/22/26	4

	Deliver Final Presentation & Demo	All	4/23/26	4/28/26	6
	Write Final Report *(milestone)*	All	**4/30/26**	**4/30/26**	1

5. Requirements

Functional Requirements:

- The device alerts the user at the scheduled Lisinopril dosage time via haptic vibration.
- The device allows medication time to be configured.
- The device allows the user to dismiss an active alert via a single dedicated button press.

Physical Requirements:

- The shell measures no greater than 55 mm (L) × 25 mm (W) × 15 mm (H).
- The shell will include attachment points on both ends with spring bar pin holes, for the silicone watch straps.
- Total system weight (device + strap) shouldn't exceed 150 grams.

Performance Requirements:

- The vibration motor will operate at a frequency between 400 Hz and 525 Hz.
- The alert triggers within 60 seconds of the scheduled medication time.
- Battery life supports 5 days of operation between charges.

Cost Requirements:

- Total unit cost won't exceed $100.

6. Performance Measures, Key Performance Parameters, and Related Objectives

TPM

TPM	Threshold (Minimum)	Objective (Target)	Test Method
Battery Life	≥ 5 days	≥ 7 days	Continuous discharge test
Alert Timing Accuracy	Within 60 seconds of scheduled time	Within 10 seconds	comparison over 24-hour period
Device Weight	≤ 150 g (with strap)	≤ 80 g	Digital scale
Enclosure Size	≤ 55 × 25 × 15 mm	≤ 52 × 23 × 13 mm	measurement

Vibration Frequency	400–525 Hz	~480 Hz	frequency analyzer

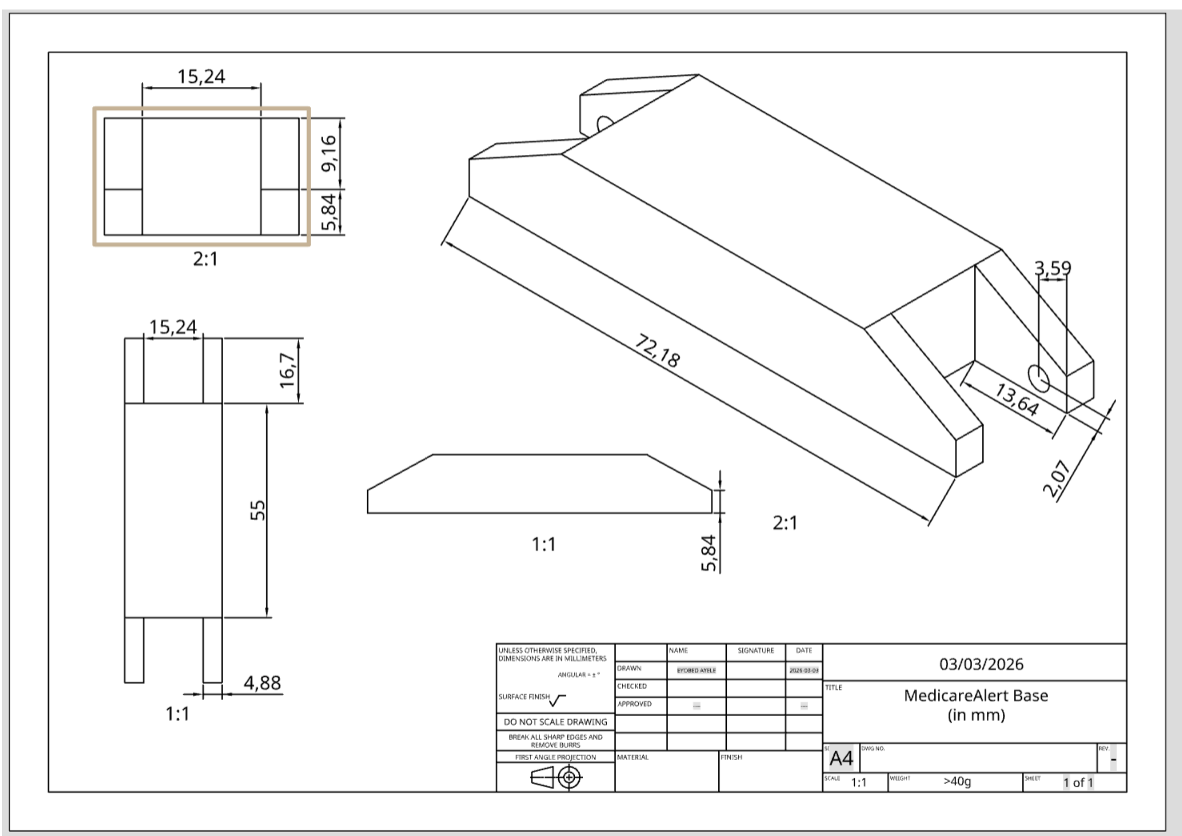
KPP

KPP	Definition	Threshold Value
KPP-1: Alert Reliability	Device triggers alert at scheduled time	100% trigger rate over 7-day test
KPP-2: Wearability	Device must be comfortable to wear daily	Good user comfort and weight ≤ 150 g
KPP-3: Battery Endurance	Device must not require daily charging	≥ 5-day battery life
KPP-4: Haptic Perceptibility	Vibration alert must be felt through the wrist	Noticeable to ≥ 90% of test participants

Related Objectives

- Design the device to be handed directly to a patient by a pharmacist.
- Ensure the enclosure is manufacturable using low-cost 3D printing for the prototype phase.
- show that the full system cost remains under $100.

8. Prototype Sketch — Dimensioned Views



9. Prototype Budget

1 | Microcontroller | Arduino Nano 33 BLE Sense Rev2 [ABX00070] | 1 | $39.70
2 | Vibration Motor | 10 mm coin vibration motor, 3 V | 1 | $3.50
3 | Transistor + Diode | 2N3904 NPN transistor + 1N4148 diode for motor drive | 1 | $1.00
4 | Push Button | 6 mm tactile switch to dismiss alert | 1 | $0.50
5 | Battery | 3.7 V Li-Po battery (≈ 500–800 mAh) | 1 | $7.00
6 | Charger Module | TP4056-based Li-Po charging board with protection | 1 | $3.00
7 | Perfboard / Mini PCB | Perfboard for wiring assembly | 1 | $3.00
8 | 3D-Printed Enclosure | Wrist shell with strap attachment points | 1 | $3.00
9 | USB C-to-Micro Cable | USB cable for programming and charging path | 1 | $3.00
10 | Miscellaneous Hardware | Wire, solder, adhesive, heat-shrink | 1 | $4.00

costs are estimates for right now. The Arduino Nano 33 BLE Sense Rev2 is $39.70 (Amazon). Final costs is going to be confirmed during later phases. Final total cost will be under the $100 unit cost requirement.